Supplementary Material

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Supplementary Methods: Search strings

A systematic literature review was conducted by searching the PubMed, Embase, Web of Science, and Cochrane Central Register of Controlled Trials databases in June 2021 for studies published in the English language since 1 January 2000. The search string included 'RAS' and 'mutation' and one of the following terms: 'Head and neck cancer', 'Head and neck squamous cell carcinoma', 'Oral cancer', 'oral squamous cell carcinoma', 'tongue', 'lips', 'nasopharyngeal'/'nasopharynx', 'pharyngeal'/'pharynx', 'laryngeal'/'larynx', 'oropharyngeal'/'oropharynx', 'Salivary gland', 'sinonasal'/'nasal'/'sinus', 'oropharyngeal'/'oropharynx', 'hypopharyngeal'/'hypopharynx', or 'tonsil'.

Figure S1: Begg's funnel plots

Potential publication bias was analyzed by Begg's funnel plots, displaying the prevalence of mutations (x-axis) versus the standard error in each study (y-axis). Each dot represents a single study. The white rectangle indicates a 95% pseudo-confidence interval, and the solid middle line indicates the overall effect from the meta-analysis.

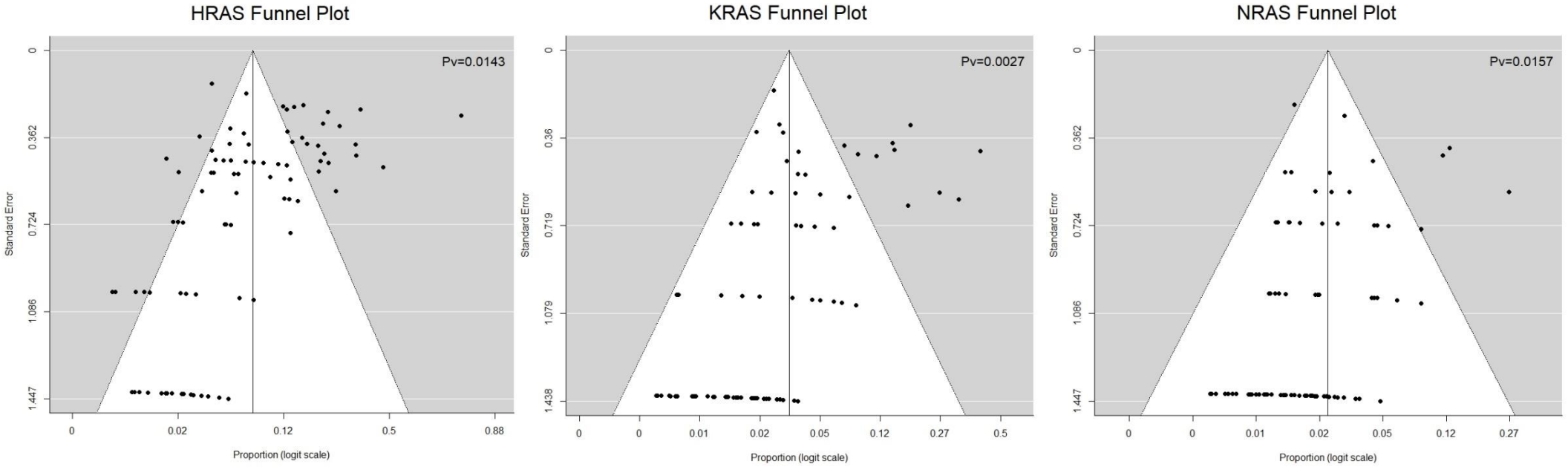

Table S1: Literature references list of studies included in the meta-analysis

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Table S2: Detailed characteristics of the studies included in the meta-analysis

	HRAS			KRAS			NRAS			HPV status	Sampling technique	Risk factors	Geographical region
Cohort	**cases**	**cohort size**	**%**	**cases**	**cohort size**	**%**	**cases**	**cohort size**	**%**				
Das et al., 2000	14	50	28.00	4	12	33.33	0	50	0.00	NP	PCR+Direct seq	T, A, B	India
Yoo et al., 2000	5	24	20.83	2	24	8.33				NP	PCR+Direct seq	NP	USA
Yoo et al., 2000	9	50	18.00							NP	PCR+Direct seq	NP	USA
Lin et al., 2002	10	28	35.71							NP	PCR+Direct seq	NP	China
Weber et al., 2003				5	89	5.62				NP	PCR+Direct seq	NP	Germany
Perrone et al., 2003	0	18	0.00							NP	PCR+Direct seq	W	Italy
Ruiz-Godoy et al., 2006				0	20	0.00				NP	PCR+Direct seq	A, S	Mexico
Sathyan et al., 2006										NP	PCR-SSCP	NP	India
Frattini et al., 2006				9	18	50.00				NP	PCR+Direct seq	W	Italy
Augello et al., 2006	11	33	33.33	2	33	6.06				NP	SSCP	NP	Italy
Russo et al., 2006	5	81	6.17	0	81	0.00	0	81	0.00	NP	PCR+Direct seq	NP	Italy
Sathyan et al., 2007	19	152	12.50	0	152	0.00	0	152	0.00	NP	PCR+Direct seq	T, A	India
Gupta et al., 2007				0	35	0.00				NP	PCR+SSCP	NP	India
Sheikh Ali et al., 2008				0	91	0.00				NP	PCR+Direct seq	NP	Japan
Chou et al., 2008				0	45	0.00				NP	PCR+Direct seq	NP	Taiwan
Bornholdt et al., 2008				8	174	4.60				NP	PCR+Direct seq	W, S, C	Denmark

Murugan et al., 2008	3	4	75.00							HPV Neg	PCR+Direct seq	NP	India
Dahse et al., 2009				1	65	1.54				NP	PCR+Direct seq	NP	Germany
Murugan et al., 2009	10	56	17.86							NP	PCR+Direct seq	T, A	Vietnam
Bruckman et al., 2010				1	42	2.38				NP	PCR+Direct seq	NP	USA
van Damme et al., 2010				1	22	4.55				NP	PCR+Direct seq	NP	Belgium
Chang et al., 2010	8	58	13.79	0	58	0.00	0	58	0.00	NP	PCR+Direct seq	NP	Taiwan
Tetsu et al., 2010	8	17	47.06	1	17	5.88	1	17	5.88	NP	PCR+Direct seq	NP	USA
Murray et al., 2010				1	92	1.09				NP	PCR+Direct seq	T, A	Greece
Popovic et al., 2010	13	60	21.67							NP	PCR+Direct seq	S	Serbia
Agarwal et al., 2011	5	120	4.17	0	120	0.00	0	120	0.00	Both	WES	T, A	USA
Stransky et al., 2011	4	74	5.41	1	74	1.35	0	74	0.00	Both	WES	T, A	USA
Cohen et al., 2011	0	37	0.00	0	37	0.00	0	37	0.00	NP	MASS SPEC and direct seq	NP	Israel
Wang et al., 2011				0	47	0.00				NP	PCR+Direct seq	NP	Taiwan
Friedland et al., 2011				0	60	0.00				Both	PCR+SSCP	S, A	Australia
Szabo et al., 2011				2	71	2.82				Both	PCR+ RFLP	S, A	Hungary
Suda et al., 2012				3	115	2.61				NP	PCR+Direct seq	T	Japan
Smilek et al. 2012				4	28	14.29				NP	qPCR	NP	Czech republic
Koumaki et al., 2012	6	86	6.98							NP	PCR+Direct seq	NP	Greece
Lopez et al., 2012				7	58	12.07				NP	PCR+Direct seq	W, T	Spain
Bissada et al., 2013				7	195	3.59				NP	PCR+ RFLP	NP	Canada

Choisea et al., 2013	1	58	1.72							HPV Pos	PCR+Direct seq	HPV, S	USA
Lechner et al., 2013	0	34	0.00	2	34	5.88	0	34	0.00	Both	MassArray platform	S, A	UK
Pickering et al., 2013	5	40	12.50	0	40	0.00	0	40	0.00	NP	WES	S, A	USA
Ho et al., 2013	5	59	8.47	0	59	0.00	1	59	1.69	NP	WES	NP	USA
Stephens et al., 2013	0	24	0.00	0	24	0.00	0	24	0.00	NP	WES	NP	UK
Liu et al., 2013	7	151	4.64	2	151	1.32	0	151	0.00	Both	WES	NP	USA
Cros et al., 2013	7	107	6.54	4	107	3.74	2	107	1.87	NP	PCR+Direct seq	NP	France
Fuji et al., 2013				0	183	0.00				NP	PCR+Direct seq	S, A	Japan
Projetti et al., 2013				2	34	5.88				NP	High Resolution Melting Analysis	W	France
Szablewski et al., 2013				12	28	42.86				NP	High Resolution Melting Analysis	W, S	France
Carvalho et al., 2013				0	94	0.00				NP	PCR+Direct seq	S, A	Brazil
Zanaruddin et al., 2013	3	107	2.80	0	107	0.00	0	107	0.00	Both	MassArray platform	S, A, B	Malaysia
Wetterskog et al., 2013	1	48	2.08	0	48	0.00	0	48	0.00	NP	PCR+Direct seq	NP	USA
Fury et al., 2013				1	18	5.56	0	18	0.00	Both	MASS SPEC and direct seq	S	USA
Chang et al., 2014	10	79	12.66	0	79	0.00	0	79	0.00	NP	Primer extension analysis	S, A, B	Taiwan
Zhang et al., 2014	1	53	1.89	6	53	11.32	6	53	11.32	Both	Targeted NGS	S, A	USA
Lin et al., 2014	0	56	0.00	2	122	1.79	1	122	1.79	NP	Targeted NGS WES, SNP array	EBV	Singapore
Ross et al., 2014	0	15	0.00	0	15	0.00	0	15	0.00	NP	Targeted NGS	NP	USA
Prigge et al., 2014				0	25	0.00				HPV Positive	PCR+Direct seq	NP	Germany
Tan et al., 2014				0	66	0.00				NP	MassArray platform	S	Singapore

Chung et al., 2014										Both	Micro Array	NP	USA
Boeckx et al., 2014				1	43	2.33				Both	High Resolution Melting Analysis and KRASstrip Assay	S	Belgium
Choisea et al., 2014	4	14	28.57							NP	PCR+Direct seq	NP	USA
Zhang et al., 2014	1	123	0.81	0	123	0.00	5	123	4.07	NP	MassArray platform	NP	china
Franchi et al., 2014				1	27	3.70				NP	PCR+Direct seq	W	Italy
Biswas et al., 2014	7	84	8.33							Both	WES	S, A, B	India
Cortelazzi et al., 2014	0	54	0.00							Both	PCR+Direct seq	NP	Italy
Pickering et al., 2014	8	39	20.51	0	39	0.00	2	39	5.13	NP	WES	NP	USA
Rampias et al., 2014	17	180	9.44							NP	PCR+Direct seq	S, A	Greece
Zhang et al., 2015				1	70	1.43				NP	SnapShot Multiplex assay	S, EBV	china
Seiwert et al., 2015	4	120	3.33	4	120	3.33	1	120	0.83	Both	Targeted NGS	S, A	USA
Vettore et al., 2015	1	60	1.67							NP	Targeted NGS	T	Singapore
Chen et al., 2015	32	345	9.28	8	345	2.32	4	345	1.15	NP	Targeted NGS	S, A, B	Taiwan
Fu et al., 2015				2	18	11.11	0	18	0.00	NP	MassArray platform	NP	Taiwan
Grunewald et al., 2015	20	84	23.81	0	84	0.00	2	84	2.38	NP	Targeted NGS	NP	Germany
Kato et al., 2015	13	117	11.11	4	117	3.42				NP	Targeted NGS	NP	USA
Choisea et al., 2015	10	29	34.48	0	29	0.00	0	29	0.00	NP	Targeted NGS	NP	USA
Er et al., 2015				0	50	0.00	0	50		NP	Targeted NGS	S, A, B	Taiwan
Fonseca et al., 2015	0	17	0.00	3	17	17.65	1	17	5.88	NP	Targeted NGS	NP	USA
Braig et al., 2016	5	46	10.87	2	46	4.35	2	46	4.35	Both	Targeted NGS	NP	Germany

Rettig et al., 2016	0	25	0.00	0	25	0.00	1	25	4.00	NP	WGS	NP	USA
Mitani et al., 2016	0	65	0.00	0	65	0.00	0	65	0.00	NP	WGS	NP	USA
Drier et al., 2016	0	10	0.00	0	19	0.00	0	10	0.00	NP	WGS	NP	USA
Shalmon et al., 2016				0	21	0.00				NP	PCR+Direct seq	NP	Israel
Wang et al., 2016	21	149	14.09	0	149	0.00	0	149	0.00	NP	Targeted NGS	NP	USA
Tinhofer et al., 2016	1	179	0.56	5	179	2.79	2	179	1.12	HPV Pos	Targeted NGS	S	Germany
Schneider et al., 2016				0	43	0.00				NP	PCR+Direct seq	NP	Germany
Choisea et al., 2016	13	38	34.21							NP	Targeted NGS	NP	USA
Dalin et al., 2016	7	31	22.58	0	31	0.00	0	31	0.00	NP	WES	NP	USA
Wu et al., 2016	8	214	3.74	18	214	8.41				NP	MassArray platform	NP	USA
Chau et al., 2016	10	213	4.69	1	213	0.47	2	213	0.94	NP	Targeted NGS	NP	USA
Bell et al., 2016	0	24	0.00	10	24	41.67	2	24	8.33	NP	MassArray platform	NP	USA
Luk et al., 2016	3	23	13.04	0	23	0.00	1	23	4.35	NP	MassArray platform	NP	Australia
Al-Hebshi et al., 2016	3	20	15.00	0	20	0.00	0	20	0.00	HPV Neg	WES	S, EBV	Yemen
Hedberg et al., 2016	0	12	0.00	1	12	8.33	1	12	8.33	NP	WES	NP	USA
Feldman et al., 2016	8	299	2.68	11	448	2.46	4	381	1.05	Both	sanger and Targeted NGS	NP	USA
Ock et al., 2016	26	71	36.62	13	71	18.31	20	71	28.17	NP	Targeted NGS	S	Korea
Oikawa et al.,2016	4	220	1.82	1	220	0.45	2	220	0.91	NP	Targeted NGS	NP	Japan
Ginkel et al., 2016	10	110	9.09	2	110	1.82	0	110	0.00	Both	Targeted NGS	S, A	Netherlands
Morris et al., 2017	9	151	5.96	2	151	1.32	2	151	1.32	Both	Targeted NGS	S, A	USA
Zhang et al., 2017	2	94	2.13	3	94	3.19	1	94	1.06	EBV pos	WES	S, EBV	China

Ali et al., 2017	1	190	0.53	3	190	1.58	4	190	2.11	Not provided	Targeted NGS	EBV	China
Upadhyay et al., 2017	3	25	12.00	0	25	0.00	0	25	0.00	NP	WES	S, B	India
Kang et al., 2017	1	18	5.56	0	18	0.00	0	18	0.00	NP	WES	T	USA
Dalin et al., 2017	1	40	2.50	0	40	0.00	0	40	0.00	NP	WES	NP	USA
Li et al., 2017	2	105	1.90	1	105	0.95	4	105	3.81	NP	WES	NP	USA/Hong Kong
Dogan et al., 2017	0	30	0.00	3	40	7.50	0	30	0.00	NP	Targeted NGS	NP	USA
Zehir et al., 2017	11	283	10.48	3	283	2.86	4	283	3.81	NP	Targeted NGS	NP	USA
Bersani et al., 2017	5	344	1.45	9	344	2.62	4	344	1.16	Both	Targeted NGS	S	Sweden
Nakagaki et al., 2017	2	47	4.26	0	47	0.00	0	47	0.00	NP	Targeted NGS	NP	Japan
Su et al., 2017	14	120	11.67							NP	WES	S, A, B	Taiwan
Robinson et al., 2017	3	29	20.00	1	29	6.67	0	29	0.00	NP	WES	NP	USA
Khoo et al., 2017	4	15	26.67	4	14	28.57	4	15	26.67	NP	Targeted NGS	NP	Australia
Saida et al., 2018	4	70	5.71	6	70	8.57	0	70	0.00	NP	Micro Array	NP	Japan
Hallani et al., 2018	8	23	34.78	0	23	0.00	0	23	0.00	NP	Targeted NGS	NP	USA
Shimura et al., 2018	23	140	16.43	0	140	0.00	0	140	0.00	NP	PCR+Direct seq	NP	Multi centered
Perdomo et al., 2018	20	180	11.11							Both	Targeted NGS	S, A, HPV	Europe and South America
Vossen etal., 2018	2	111	1.80	0	111	0.00	1	111	0.90	HPV neg	Targeted NGS	S, A	Netherlands
Dubot et al.,2018	2	122	1.64	2	122	1.64	1	122	0.82	Both	PCR+Direct seq	S, A, HPV	France
Mirghani et al., 2018	0	62	0.00	2	62	3.23	0	62	0.00	HPV Pos	Targeted NGS	S, T, HPV	France

Nakagaki et al., 2018	4	80	5.00	0	80	0.00	0	80	0.00	Both	Targeted NGS	NP	Japan
Li et al., 2018	34	168	20.24							NP	Amplification-refractory mutation system	NP	China
Batta et al., 2019	8	46	17.39	0	46	0.00	0	46	0.00	NP	Targeted NGS	T	India
Akagi et al., 2019				1	85	1.18				NP	PCR+Direct seq	NP	Japan
Chung et al., 2019	0	33	0.00	1	33	3.03	0	33	0.00	NP	Targeted NGS	S, EBV	Taiwan
Reder et al., 2019	5	24	41.67	1	24	8.33	1	24	8.33	NP	Targeted NGS	S, A	Germany
Stanek et al., 2019				7	55	12.73	2	54	3.70	Both	PCR- mutation detection kit	NP	Czech republic
Nakaguro et al., 2019	7	21	33.33							NP	PCR+Direct seq	NP	Japan
Wang et al., 2019				30	80	37.50				NP	PCR+Direct seq	S	China
Haft et al., 2019	2	46	4.35	0	46	0.00	0	46	0.00	HPV Pos	WES	S, A, HPV	USA
Smith et al., 2019	0	21	0.00	1	21	4.76	0	21	0.00	NP	Targeted NGS	S	USA
Westbrook et al., 2019				1	23	4.35	0	23	0.00	NP	Targeted NGS	S	USA
Samstein et al., 2019	5	139	3.60	3	139	2.16	3	139	2.16	NP	Targeted NGS	NP	USA
Priestley et al., 2019	2	42	4.76	0	42	0.00	0	42	0.00	NP	WGS	NP	Netherlands
Pérez Sayáns et al 2019	33	528	6.25	9	528	1.70	14	528	2.65	Both	Targeted NGS, WGS, WES	S	Multi centered
Kobyashi et al., 2019	7	284	2.46							Both	PCR+Direct seq	S, A	Japan
Dogan et al., 2019	6	25	24.00	0	25	0.00	0	25	0.00	NP	Targeted NGS	S	USA
Gauthaman et al., 2020				34	56	60.71				NP	PCR-RFLP	NP	India
Kyurkchiyan et al., 2020	3	57	5.26	2	57	3.51	1	57	1.75	HPV Neg	Targeted NGS	NP	Bulgaria

Morita et al., 2020	2	101	1.98	7	101	6.93	0	101	0.00	NP	SnapShot Multiplex assay	NP	Japan
Morfouace et al., 2020	0	14	0.00	0	14	0.00				NP	Targeted NGS	NP	France
Kawamura et al., 2020	0	20	0.00	0	20	0.00	0	20	0.00	NP	PCR-SSCP	NP	Japan
Jayaprakash et al., 2020	6	28	21.43	0	28	0.00	0	28	0.00	NP	Targeted NGS	NP	India
Leblanc et al., 2020	4	115	3.48	2	115	1.74	0	115	0.00	Both	High Resolution Melting Analysis and sanger	NP	France
Mueller et al., 2020	14	63	22.22	3	63	4.76	0	63	0.00	NP	Targeted NGS	S	Australia
Kim et al., 2020	4	42	9.52	0	42	0.00	0	42	0.00	NP	Targeted NGS	S	Korea
Hsieh et al., 2020	9	33	27.27				8	33	24.24	NP	PCR+Direct seq	NP	Taiwan
ORCA ICGC	21	178	11.80	6	178	3.37	2	178	1.12	NP	targeted NGS, WGS	S, A, B	India
Masato et al., 2021	66	83	79.52	0	83	0.00	0	83	0.00	NP	PCR+Direct seq	NP	Japan
Sanchez-Fernandez et al., 2021	0	48	0.00	7	48	14.58	2	48	4.17	NP	Targeted NGS	S, A, W	Spain
AACR GENIE V9.1	55	1636	3.36	37	1636	2.26	20	1636	1.22	NP	WES	S	Multi centered
Reder et al., 2021	9	56	16.07	8	56	14.29	7	56	12.50	Both	Targeted NGS	S, A	Germany
Patel et al., 2021	4	30	13.33	0	30	0.00	0	30	0.00	NP	WES	T	India

NP-not provided, HPV pos- human papillomavirus positive, HPV neg- human papillomavirus negative, S-smoking T-Tabacco, A-Alcohol, B-betel chewing, W wood/leather dust, C- other chemical exposure, HPV-. human papillomavirus, EBV- Epstein–Barr virus

Table S3: Detailed anatomical site data

	HRAS															
	Oral cavity		**Salivary gland**		**Sinonasal**		**Nasopharynx**		**Oropharynx**		**Hypopharynx**		**Larynx**		**Other**	
cohort	mutated	total	mutated	total	mutated	total	mutated	total	mutated	total	mutated	total	mutated	total	mutated	total
Das et al., 2000	14	50														
Yoo et al., 2000			5	24												
Yoo et al., 2000			9	50												
Weber et al., 2003																
Perrone et al., 2003					0	18										
Ruiz-Godoy et al., 2006																
Sathyan et al., 2006																
Frattini et al., 2006																
Augello et al., 2006			11	33												
Sathyan et al., 2007	19	152														
Gupta et al., 2007	0	20					0	1					0	14		
Sheikh Ali et al., 2008																
Chou et al., 2008																
Bornholdt et al., 2008																
Dahse et al., 2009																
Murugan et al., 2009	10	56														
Bruckman et al., 2010																
van Damme et al., 2010																
Chang et al., 2010	8	58														
Tetsu et al., 2010	7	8	0	3	0	1									3	5
Murray et al., 2010																
Agarwal et al., 2011	4	75							1	21	0	9	0	13		
Stransky et al., 2011	3	51			0	2			0	15	0	9	1	15		
Cohen et al., 2011	0	37														
Trivedi et al., 2011																
Wang et al., 2011																
Friedland et al., 2011																

Szabo et al., 2011																
Szanyi et al., 2011																
Suda et al., 2012																
Koumaki et al., 2012	3	86														
Lopez et al., 2012 and Garcia-Inclan et al., 2012																
Smilek et al., 2012																
Bissada et al., 2013																
Choisea et al., 2013									1	58						
Lechner et al., 2013									0	34						
Pickering et al., 2013	5	40														
Ho et al., 2013			5	59												
Stephens et al., 2013			0	24												
Liu et al., 2013																
Cros et al., 2013			7	107												
Fuji et al., 2013																
Projetti et al., 2013																
Szablewski et al., 2013																
Carvalho et al., 2013																
Zanaruddin et al., 2013	3	107														
Wetterskog et al., 2013																
Fury et al., 2013																
Chang et al., 2014	10	79														
Zhang et al., 2014																
Lin et al., 2014							0	56								
Ross et al., 2014			0	15												
Al Rawi et al., 2014																
Prigge et al., 2014																
Tan et al., 2014																
Chung et al., 2014																
Boeckx et al., 2014																
Choisea et al., 2014			4	14												
Lin et al., 2014																

Zhang et al., 2014							1	123								
Franchi et al., 2014																
Zhang et al., 2015																
Seiwert et al., 2015																
Vettore et al., 2015	1	60														
Chen et al., 2015	32	345														
Fu et al., 2015																
Grunewald et al., 2015			20	84												
Kato et al., 2015			13	117												
Choisea et al., 2015			10	29												
Braig et al., 2016	3	12			0	2			1	19	1	9	0	4		
Rettig et al., 2016			0	25												
Mitani et al., 2016			0	65												
Drier et al., 2016			0	10												
Shalmon et al., 2016																
Wang et al., 2016			2	149												
Tinhofer et al., 2016																
Kucuk et al., 2016																
Schneider et al., 2016																
Choisea et al., 2016			13	38												
Dalin et al., 2016			7	31												
Udager et al., 2016																
Wu et al., 2016	4	56							0	76			4	82		
Chau et al., 2016	3	60			0	5	0	7	3	97	0	6	1	24	3	14
Bell et al., 2016															0	24
Luk et al., 2016			3	23												
Al-Hebshi et al., 2016																
Hedberg et al., 2016	0	6									0	3	0	4		
Chuerduangphui et al., 2017																
Morris et al., 2017	1	26	4	32	1	15	0	9	0	23	0	2	0	8	3	36
Yue et al., 2017																

Abdolkarim Moazeni-Roodi et al., 2017																
Zhang et al., 2017							1	94								
Ali et al., 2017																
Upadhyay et al., 2017	3	24														
kang et al., 2017			1	18												
Dalin et al., 2017			1	40												
Li et al., 2017							1	105								
Dogan et al., 2017					0	30										
Krishna et al., 2018																
Lin et al., 2018																
Saida et al., 2018			4	70												
Hallani et al., 2018			8	23												
Shimura et al., 2018			23	140												
Perdomo et al., 2018																
Vossen etal., 2018																
Batta et al., 2019	8	39													0	7
Akagi et al., 2019																
Chung et al., 2019																
Reder et al., 2019									5	12						
Stanek et al., 2019																
Urano et al., 2019																
Nakaguro et al., 2019			7	21												
Wang et al., 2019																
Reder et al., 2019									0	12						
Gauthaman et al., 2020																
Kyurkchiyan et al., 2020																
ORCA ICGC													3	57		
Sasaki et al., 2020	6	12									7	20	0	7	4	12
Morita et al., 2020			2	101												
Morfouace et al., 2020					0	3	0	10							0	1
Kawamura et al., 2020	0	20														
Pérez Sayáns et al 2019	28	303							1	82	0	10	1	117		

Masato et al., 2021																
Sanchez-Fernandez et al., 2021					0	48										
AACR GENIE V9.0	21	468	38	791	3	80	0	82	4	306	0	29	4	98	0	21
Dubot et al., 2018	0	61							1	22	0	17	1	22		
Zehir et al., 2017	4	59	1	9	1	12	0	17	0	44	0	3	0	16	0	26
Zehir et al., 2017 SG			5	105												
Bersani et al., 2017									5	325					0	19
Biswas et al., 2014	7	84														
Haft et al., 2019									2	46						
Jayaprakash et al., 2019	6	28														
Mirghani et al., 2018									0	62						
Nakagaki et al., 2017	2	47														
Ock et al., 2016																
Oikawa et al., 2016	6	220														
Smith et al., 2019													0	21		
Su et al., 2017																
Westbrook et al., 2019																
Samstein et al., 2019	2	56			1	6	0	14	2	31	0	8	0	14	0	9
Robinson et al., 2017	0	8			0	1			1	2			0	1	1	2
Robinson et al., 2017			1	14												
Priestley et al., 2019	0	12	1	19	0	10	0	1	0	3	0	5	1	2	0	1
Popovic et al., 2010	13	60													8	39
Pickering et al., 2014																
Leblanc et al., 2020																
Mueller et al., 2020			14	63												
Kim et al., 2020			4	42												
Hsieh et al., 2020			9	33												
Khoo et al., 2017			4	15												
Nakagaki et al., 2018																
Reder et al., 2021									9	56						
Patel et al., 2021	4	30														
Fonseca et al., 2015			0	17												
Li et al., 2018							34	168								

Lin et al., 2002													10	28		
Russo et al., 2006													5	81		
Kobyashi et al., 2019	6	153							1	32	0	74	0	25		
Dogan et al., 2019			6	25												
Ginkel et al., 2016	6	34							3	37	1	16	0	18	1	14
Rampias et al., 2014									3	13	0	8	7	82	0	3

	KRAS															
	Oral cavity		**Salivary gland**		**Sinonasal**		**Nasopharynx**		**Oropharynx**		**Hypopharynx**		**Larynx**		**Other**	
Cohorts	mutated	total	mutated	total	mutated	total	mutated	total	mutated	total	mutated	total	mutated	total	mutated	total
Das et al., 2000	4	12														
Yoo et al., 2000			2	24												
Yoo et al., 2000																
Weber et al., 2003	1	13							1	33	2	18	1	25		
Perrone et al., 2003																
Ruiz-Godoy et al., 2006													0	20		
Sathyan et al., 2006																
Frattini et al., 2006					9	18										
Augello et al., 2006			2	33												
Sathyan et al., 2007	0	152														
Gupta et al., 2007	0	20					0	1					0	14		
Sheikh Ali et al., 2008	0	51							0	9	0	14	0	17		
Chou et al., 2008							0	45								
Bornholdt et al., 2008					8	174										
Dahse et al., 2009			1	65												
Murugan et al., 2009																
Bruckman et al., 2010	1	42														
van Damme et al., 2010									1	22						
Chang et al., 2010	0	58														
Tetsu et al., 2010	1	8	0	3	0	1									0	5
Murray et al., 2010																

Agarwal et al., 2011	0	75							0	21	0	9	0	13		
Stransky et al., 2011	0	51			0	2			0	15	1	9	0	15		
Cohen et al., 2011	0	37														
Trivedi et al., 2011																
Wang et al., 2011	0	47														
Friedland et al., 2011									0	20						
Szabo et al., 2011																
Szanyi et al., 2011																
Suda et al., 2012	2	31			0	11			1	25	0	25	0	23		
Koumaki et al., 2012																
Lopez et al., 2012 and Garcia-Inclan et al., 2012					7	58										
Smilek et al., 2012																
Bissada et al., 2013	0	21							4	123	0	11	3	29	0	11
Choisea et al., 2013																
Lechner et al., 2013					`				2	34						
Pickering et al., 2013	0	40														
Ho et al., 2013			0	59												
Stephens et al., 2013			0	24												
Liu et al., 2013																
Cros et al., 2013			4	107												
Fuji et al., 2013									0	71	0	83	0	51		
Projetti et al., 2013					2	34										
Szablewski et al., 2013					12	28										
Carvalho et al., 2013	0	66							0	12	0	2	0	14		
Zanaruddin et al., 2013	0	107														
Wetterskog et al., 2013																
Fury et al., 2013	1	1			0	1	0	1	0	14	0	1				
Chang et al., 2014	0	79														
Zhang et al., 2014																
Lin et al., 2014							1	56								
Ross et al., 2014			0	15												
Al Rawi et al., 2014																

Prigge et al., 2014	0	3							0	21	0	1				
Tan et al., 2014	0	66														
Chung et al., 2014																
Boeckx et al., 2014																
Choisea et al., 2014																
Lin et al., 2014							1	66								
Zhang et al., 2014							0	123								
Franchi et al., 2014					1	27										
Zhang et al., 2015							1	70								
Seiwert et al., 2015																
Vettore et al., 2015																
Chen et al., 2015	8	345														
Fu et al., 2015			2	18												
Grunewald et al., 2015			0	84												
Kato et al., 2015																
Choisea et al., 2015			0	29												
Braig et al., 2016	0	12			0	2			0	19	1	9	1	4		
Rettig et al., 2016			0	25												
Mitani et al., 2016			0	65												
Drier et al., 2016			0	10												
Shalmon et al., 2016			0	21												
Wang et al., 2016			0	149												
Tinhofer et al., 2016																
Kucuk et al., 2016	0	26														
Schneider et al., 2016			0	43												
Choisea et al., 2016																
Dalin et al., 2016			0	31												
Udager et al., 2016					56	56										
Wu et al., 2016	1	56							8	76			9	82		
Chau et al., 2016	1	60			0	5	0	7	0	97	0	6	0	24	0	14
Bell et al., 2016															10	24
Luk et al., 2016			0	23												

Al-Hebshi et al., 2016																
Hedberg et al., 2016	0	6									0	3	1	4		
Chuerduangphui et al., 2017																
Morris et al., 2017	1	26	4	32	1	15	0	9	0	23	0	2	0	8	3	36
Yue et al., 2017																
Abdolkarim Moazeni-Roodi et al., 2017																
Zhang et al., 2017							3	94								
Ali et al., 2017																
Upadhyay et al., 2017	0	25														
kang et al., 2017			0	18												
Dalin et al., 2017			0	40												
Li et al., 2017							1	105								
Dogan et al., 2017					3	30										
Krishna et al., 2018																
Lin et al., 2018																
Saida et al., 2018			6	70												
Hallani et al., 2018			0	23												
Shimura et al., 2018			0	140												
Perdomo et al., 2018																
Vossen etal., 2018																
Batta et al., 2019	0	39													0	7
Akagi et al., 2019																
Chung et al., 2019																
Reder et al., 2019									0	12						
Stanek et al., 2019																
Urano et al., 2019																
Nakaguro et al., 2019																
Wang et al., 2019					30	80										
Reder et al., 2019									1	12						
Gauthaman et al., 2020	25	44							2	3	4	6	0	0	3	4
Kyurkchiyan et al., 2020													2	57		

ORCA ICGC																
Sasaki et al., 2020	1	12									11	20	2	7	4	12
Morita et al., 2020			7	101												
Morfouace et al., 2020					0	3	0	10							0	1
Kawamura et al., 2020	0	20														
Pérez Sayáns et al 2019	1	303							0	82	0	10	0	117		
Masato et al., 2021																
Sanchez-Fernandez et al., 2021															7	48
AACR GENIE V9.0	8	468	8	791	0	80	2	82	4	306	0	29	4	98	0	21
Dubot et al., 2018	0	61							2	22	0	17	0	22		
Zehir et al., 2017	1	59	0	9	0	12	0	17	0	52	0	3	1	16	0	18
Zehir et al., 2017 SG			1	105												
Bersani et al., 2017									9	325					0	19
Biswas et al., 2014																
Haft et al., 2019									0	46						
Jayaprakash et al., 2019	0	28														
Mirghani et al., 2018									2	62						
Nakagaki et al., 2017	0	47														
Ock et al., 2016																
Oikawa et al., 2016	1	220														
Smith et al., 2019																
Su et al., 2017													1	21		
Westbrook et al., 2019	0	6					0	1	0	7	0	2	1	7		
Samstein et al., 2019	1	56			0	6	0	14	1	33	0	8	1	14	0	9
Robinson et al., 2017	0	8			0	1			0	2			0	1	1	2
Robinson et al., 2017			0	14												
Priestley et al., 2019	0	12	0	19	0	10	0	1	0	3	0	5	0	2	0	1
Popovic et al., 2010															0	39
Pickering et al., 2014																
Leblanc et al., 2020																
Mueller et al., 2020			3	63												
Kim et al., 2020			0	42												
Hsieh et al., 2020																

Khoo et al., 2017			4	15												
Nakagaki et al., 2018																
Reder et al., 2021									8	56						
Patel et al., 2021	0	30														
Fonseca et al., 2015			3	17												
Li et al., 2018																
Lin et al., 2002																
Russo et al., 2006													0	81		
Kobyashi et al., 2019																
Dogan et al., 2019			0	25												
Ginkel et al., 2016									1	37					1	14
Rampias et al., 2014																

	NRAS															
	Oral cavity		**Salivary gland**		**Sinonasal**		**Nasopharynx**		**Oropharynx**		**Hypopharynx**		**Larynx**		**Other**	
Cohorts	mutated	total	mutated	total	mutated	total	mutated	total	mutated	total	mutated	total	mutated	total	mutated	total
Das et al., 2000																
Yoo et al., 2000																
Yoo et al., 2000																
Weber et al., 2003																
Perrone et al., 2003																
Ruiz-Godoy et al., 2006																
Sathyan et al., 2006																
Frattini et al., 2006																
Augello et al., 2006																
Sathyan et al., 2007	0	152														
Gupta et al., 2007	0	20					0	1					0	14		
Sheikh Ali et al., 2008																
Chou et al., 2008																
Bornholdt et al., 2008																
Dahse et al., 2009																

Murugan et al., 2009																
Bruckman et al., 2010																
van Damme et al., 2010																
Chang et al., 2010	0	58														
Tetsu et al., 2010	0	8	1	3	0	1									0	5
Murray et al., 2010																
Agarwal et al., 2011	0	75							0	21	0	9	0	13		
Stransky et al., 2011	0	51			0	2			0	15	0	9	0	15		
Cohen et al., 2011	0	37														
Trivedi et al., 2011																
Wang et al., 2011																
Friedland et al., 2011																
Szabo et al., 2011																
Szanyi et al., 2011																
Suda et al., 2012																
Koumaki et al., 2012																
Lopez et al., 2012 and Garcia-Inclan et al., 2012																
Smilek et al., 2012																
Bissada et al., 2013																
Choisea et al., 2013																
Lechner et al., 2013									0	34						
Pickering et al., 2013	0	40														
Ho et al., 2013			1	59												
Stephens et al., 2013			0	24												
Liu et al., 2013																
Cros et al., 2013			2	107												
Fuji et al., 2013																
Projetti et al., 2013																
Szablewski et al., 2013																
Carvalho et al., 2013																
Zanaruddin et al., 2013	0	107														

Wetterskog et al., 2013																
Fury et al., 2013	1	1			0	1	0	1	0	14	0	1				
Chang et al., 2014	0	79														
Zhang et al., 2014																
Lin et al., 2014							1	56								
Ross et al., 2014			0	15												
Al Rawi et al., 2014																
Prigge et al., 2014																
Tan et al., 2014																
Chung et al., 2014																
Boeckx et al., 2014																
Choisea et al., 2014																
Lin et al., 2014							0	66								
Zhang et al., 2014							5	123								
Franchi et al., 2014																
Zhang et al., 2015																
Seiwert et al., 2015																
Vettore et al., 2015																
Chen et al., 2015																
Fu et al., 2015			0	18												
Grunewald et al., 2015			2	84												
Kato et al., 2015																
Choisea et al., 2015			0	29												
Braig et al., 2016	1	12			0	2			0	19	0	9	1	4		
Rettig et al., 2016			1	25												
Mitani et al., 2016			0	65												
Drier et al., 2016			0	10												
Shalmon et al., 2016																
Wang et al., 2016			0	149												
Tinhofer et al., 2016																
Kucuk et al., 2016																
Schneider et al., 2016																

Choisea et al., 2016																
Dalin et al., 2016			0	31												
Udager et al., 2016																
Wu et al., 2016																
Chau et al., 2016	2	60			0	5	0	7	0	97	0	6	0	24	0	14
Bell et al., 2016															2	24
Luk et al., 2016			1	23												
Al-Hebshi et al., 2016																
Hedberg et al., 2016	1	6									0	3	0	4		
Chuerduangphui et al., 2017																
Morris et al., 2017	1	26	0	32	0	15	1	9	0	23	0	2	0	8	0	36
Yue et al., 2017																
Abdolkarim Moazeni-Roodi et al., 2017																
Zhang et al., 2017							1	94								
Ali et al., 2017																
Upadhyay et al., 2017	0	24														
kang et al., 2017			0	18												
Dalin et al., 2017			0	40												
Li et al., 2017							3	105								
Dogan et al., 2017					0	30										
Krishna et al., 2018																
Lin et al., 2018																
Saida et al., 2018			0	70												
Perdomo et al., 2018																
Vossen etal., 2018																
Hallani et al., 2018			0	23												
Shimura et al., 2018			0	140												
Batta et al., 2019	0	39													0	7
Akagi et al., 2019																
Chung et al., 2019																
Reder et al., 2019																

Stanek et al., 2019									1	12						
Urano et al., 2019																
Nakaguro et al., 2019																
Wang et al., 2019																
Reder et al., 2019									0	12						
Gauthaman et al., 2020																
Kyurkchiyan et al., 2020													1	57		
ORCA ICGC																
Sasaki et al., 2020																
Morita et al., 2020			0	101												
Morfouace et al., 2020					0	3	0	10							0	1
Kawamura et al., 2020	0	20														
Pérez Sayáns et al 2019	0	303							0	82	0	10	1	117		
Masato et al., 2021																
Sanchez-Fernandez et al., 2021															2	48
AACR GENIE V9.0	5	468	5	791	1	80	1	82	3	306	0	29	0	98	0	21
Dubot et al., 2018	1	61							0	22	0	17	0	22		
Zehir et al., 2017	1	59	0	9	0	12	1	17	0	52	0	3	0	16	1	18
Zehir et al., 2017 SG			1	105												
Bersani et al., 2017									4	325					0	19
Biswas et al., 2014																
Haft et al., 2019									0	46						
Jayaprakash et al., 2019	0	28														
Mirghani et al., 2018									0	62						
Nakagaki et al., 2017	0	47														
Ock et al., 2016																
Oikawa et al., 2016	2	220														
Smith et al., 2019													0	21		
Su et al., 2017																
Westbrook et al., 2019	0	6					0	1	0	7	0	2	0	7		
Samstein et al., 2019	1	56			0	6	1	14	1	33	0	8	0	14	0	9

Robinson et al., 2017	0	8			0	1			0	2			0	1	0	2
Robinson et al., 2017			0	14												
Priestley et al., 2019	0	12	0	19	0	10	0	1	0	3	0	5	0	2	0	1
Popovic et al., 2010															2	39
Pickering et al., 2014																
Leblanc et al., 2020																
Mueller et al., 2020			0	63												
Kim et al., 2020			0	42												
Hsieh et al., 2020			8	33												
Khoo et al., 2017			4	15												
Nakagaki et al., 2018																
Reder et al., 2021									7	56						
Patel et al., 2021	0	30														
Fonseca et al., 2015			1	17												
Li et al., 2018																
Lin et al., 2002																
Russo et al., 2006													0	81		
Kobyashi et al., 2019																
Dogan et al., 2019			0	25												
Ginkel et al., 2016																
Rampias et al., 2014																